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نظري

كلية الهندسة المعلوماتية

السنة الثانية

Operations on signals

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الاتصالات الرقمية



Key points in the lecture:

- Amplitude scaling
- Amplitude shifting
- Addition of signals
- Multiplication of signals

Amplitude Scaling

Amplitude shifting of a unit step signal $u(t)$ changes its peak value from 1 to a new amplitude A

How?

- By multiplying the signal by a constant: $y(t) = A \cdot u(t)$.

This results:

- A step signal with Amplitude A for $t > 0$
- while remaining 0 for $t < 0$

Changing its intensity rather than position.

Amplification:

If $a > 1$, the amplitude of the signal is increased.

Attenuation:

If $a < 1$, the amplitude of the signal is decreased.

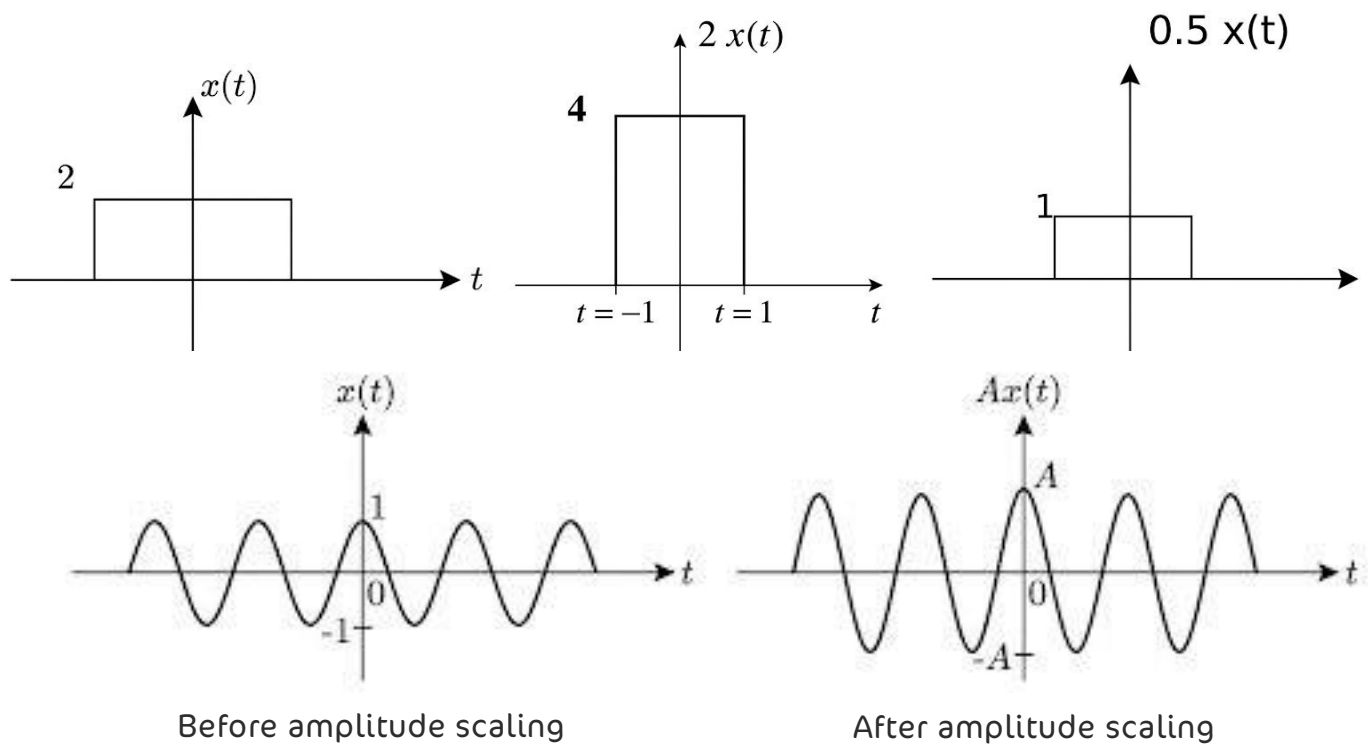
Inversion:

If $a < 0$, the signal is flipped across the time axis (phase inversion).

Examples:

- Amplifying a Wave: $y(t) = 2 x(t)$ doubles the amplitude
- Attenuating a Wave: $y(t) = 0.5 x(t)$ reduces the amplitude by half.





Amplification ($a > 1$):

Increases the signal's peak-to-peak voltage or intensity,
Making weak signals usable for further processing

Attenuation ($a < 1$):

Decreases signal amplitude to prevent overloading or clipping in devices

Like: amplifiers.

Signal Normalization:

Adjusts multiple signals to common range for easier comparison or processing.

Analog Computation:

Used in analog computing to scale input variables.

Waveform Control:

Modifies the amplitude in modulation techniques

Like amplitude modulation

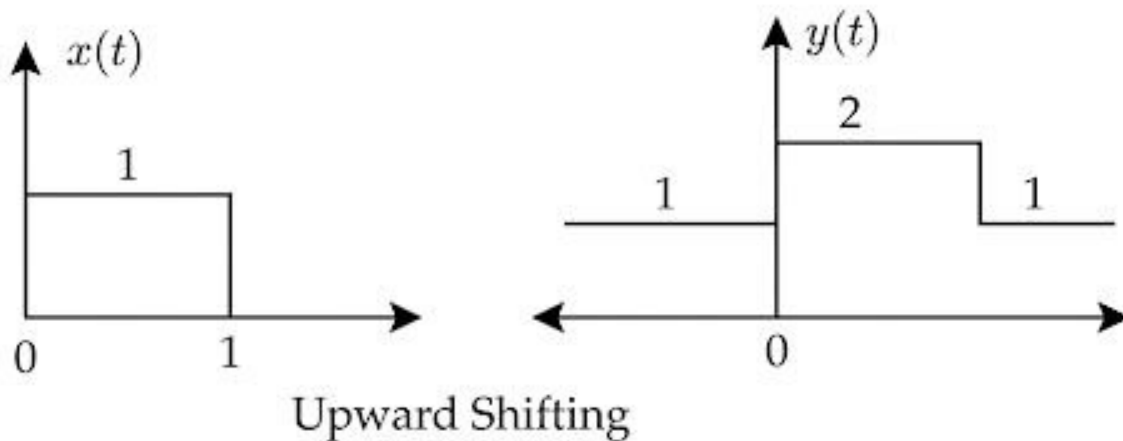
To carry information without changing the time-domain

Amplitude shifting

Amplitude shifting of a signal: is a vertical transformation that adds a constant K to the signal's amplitude, resulting in: $y(t) = x(t) + k$.

- Positive K shifts the waveform upward.
- Negative K shifts the waveform downward.

This operation changes the DC (Direct current) offset without affecting the original signal's shape or time-domain positioning.



Note:

Amplitude shifting:

The values don't change with respect to time, only change with respect to amplitude.

Time shifting:

The values don't change with respect to amplitude, only change with respect to time.



Stay
focus

Preparing Signals for Unipolar Components:

Many electronic components, **such as** single-supply amplifiers or digital converters (ADC), only operate with positive voltages

A bipolar signal (AC signal) can be shifted upward to ensure the entire waveform stays above zero.

Fitting Signal Processing Ranges:

Electronic components (like ADCs-Analog to Digital Converters) often have limited input voltage ranges, such as 0V to 5V.

If a signal is centered around a higher voltage (e.g., 10V),

It must be shifted downward to fit within the operating window of the circuit without distortion.

AC: Alternating current

ADC: Analog to digital converter

DAC: Digital to analog converter

In the early days of the internet, we used a regular modem.

Modem: Modulator Demodulator.

Addition of Signals

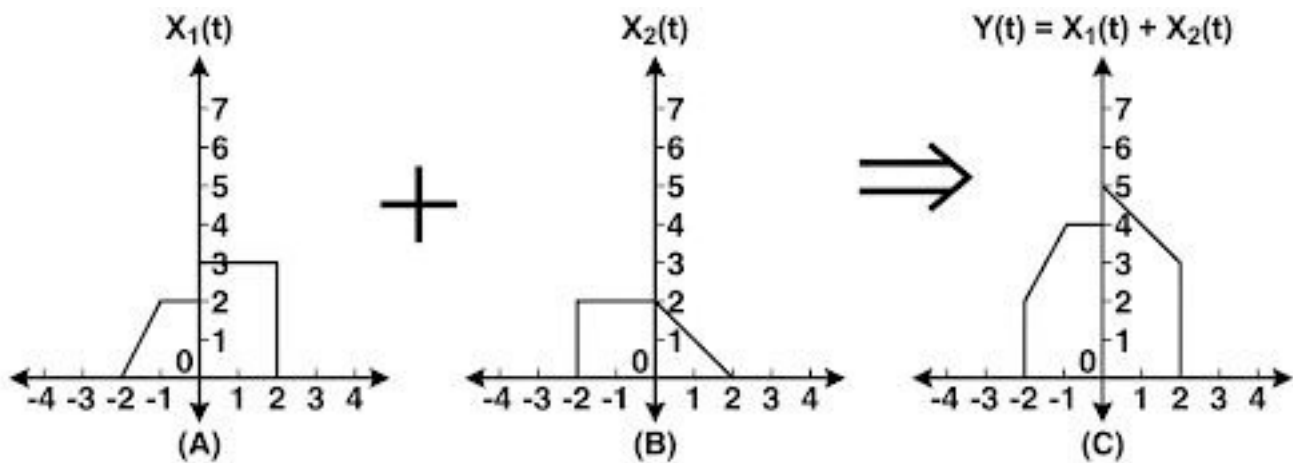
The sum of signals, $x(t) = x_1(t) + x_2(t)$ is a new signal formed by adding the amplitudes of two or more individual signals at each corresponding instant in time.

This process, known as superposition, is used to combine waveforms for analysis, **such as** adding audio signals for a 6dB increase (if fully correlated) or calculating periodic signal combinations.

Why?

Adding two signals, or signal summation, combines their respective amplitudes at every instant in time, **commonly used to:**

- Mix information.
- Create complex waveforms.
- Strengthen signals.
- For audio mixing (combining voice and music).
- Enhancing signal strength (radio transmitters).
- Analyzing interference (beating frequencies).



- $x_1(t), x_2(t)$: are continuous signals.
- Addition of signals based on value by value.
- It starts from the left, put the result and connect the signals.
- The shape is diagonal because we summed the values in this case at (-1) and (-2) and put the result and connected the signals because they are **continuous**.

Home work (1):

Subtraction of signals ... How & Why?

“In digital communication, every bit carries the power to change the world” 💪 🔥 🔥

Multiplication of signals

Signal multiplication is a fundamental signal processing operation involving point-by-point multiplication of two signals' amplitudes at each instant $y(t) = x_1(t) x_2(t)$ for continuous.

$y[n] = x_1[n] x_2[n]$ for discrete.

It is key for AM modulation, envelope detection, and shaping. where it effectively shapes one signal with the other.

In radio:

The majority of radio stations previously operated on amplitude modulation technology (AM).

The channel broadcasts a song which is a continuous signal called carrier. On this signal, it carries the song, it takes the signal and the song and performs a multiplication process for them.

The song has amplitude and a phase that only modulates the amplitude of the carrier

The carrier multiplies it with the song signal

The song signal adjusts the amplitude of the carrier

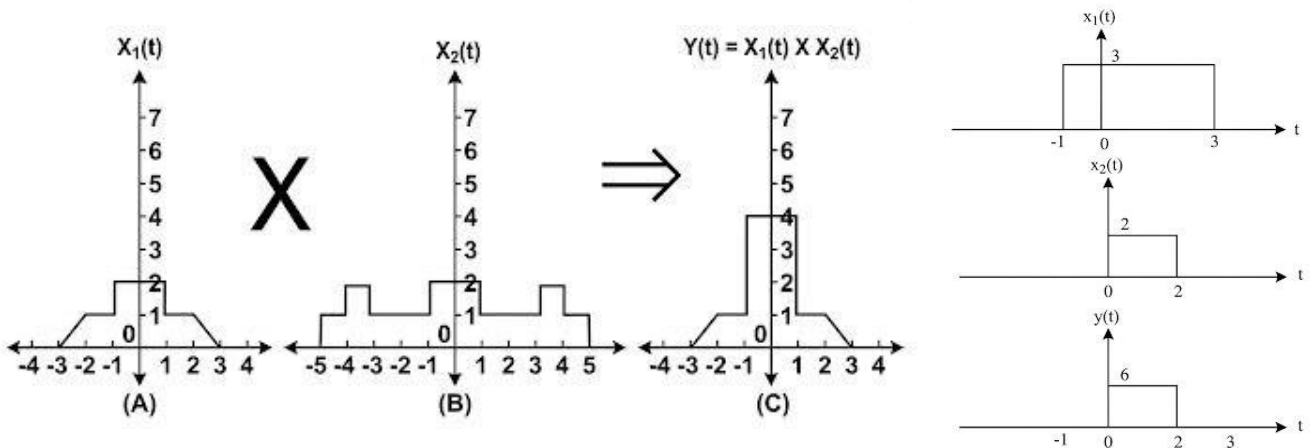
The radio picks up the signal, which is the carrier, and the song takes the signal and excludes the carrier, so only the song remains

Common Applications:

- **Communication Systems:** Used in amplitude modulation (AM) to shift the baseband message signal to a higher frequency carrier, according to.
- **Signal Processing:** Used for filtering, shaping wave envelopes, and to measure phase differences between signals.
- **Audio Modulation:** Used to create effects like ring modulation.

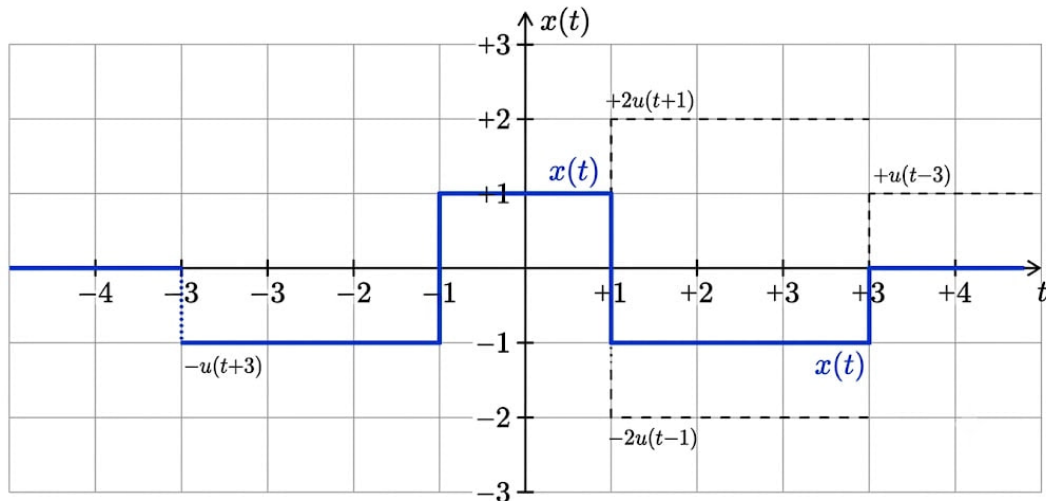
Application: Crucial in modulation (e.g., Amplitude Modulation - AM), where a baseband signal is multiplied by a carrier wave to shift its frequency spectrum.

Multiplication of Signals



Exercises

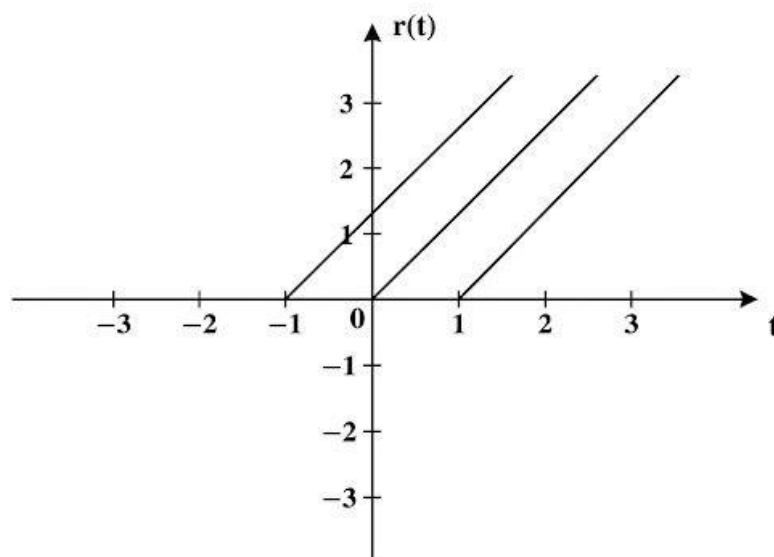
1) $x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$



Solution steps:

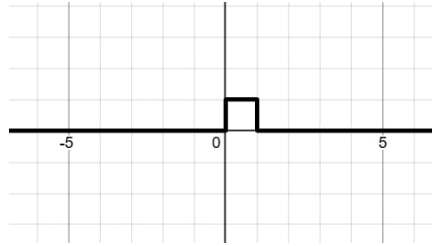
- 1) We start from the expression (continuous)
- 2) Select the signal (u)
- 3) Determine the points I want
- 4) Addition process
- 5) Clear the remaining signal

Example of Ramp signal:



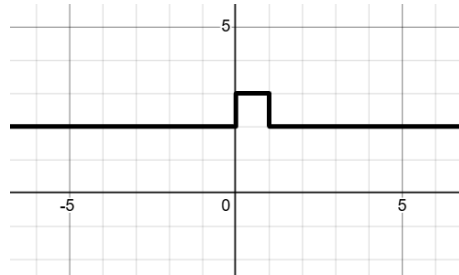
$r(t+1)$: Advance
 $r(t-1)$: Delay

$$x(t) = \begin{cases} 0 & t < 0 \\ 1 & 0 \leq t \leq 1 \\ 0 & t > 1 \end{cases}$$



$$Y(t) = x(t) + 2$$

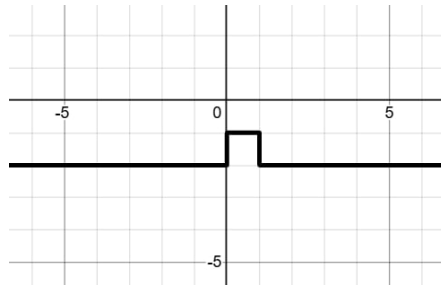
$$Y(t) = \begin{cases} 2 & t < 0 \\ 3 & 0 \leq t \leq 1 \\ 2 & t > 1 \end{cases}$$



Upward

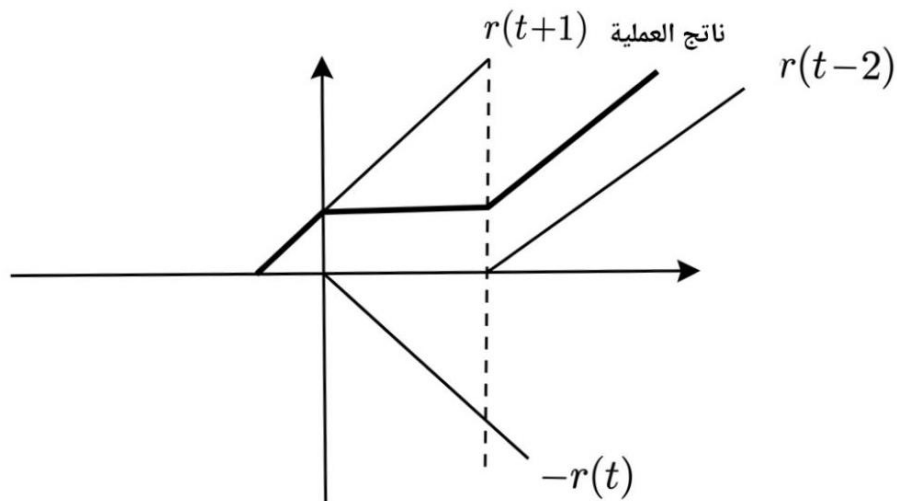
$$Y(t) = x(t) - 2$$

$$Y(t) = \begin{cases} -2 & t < 0 \\ -1 & 0 \leq t \leq 1 \\ -2 & t > 1 \end{cases}$$



Downward

$$2) y(t) = r(t+1) - r(t) + r(t-2)$$



وظيفة: ما التغيير الذي يطرأ إذا أجرينا التعديل التالي على المعادلة السابقة؟

$$y(t) = r(t+1) - 2r(t) + 3r(t-2)$$

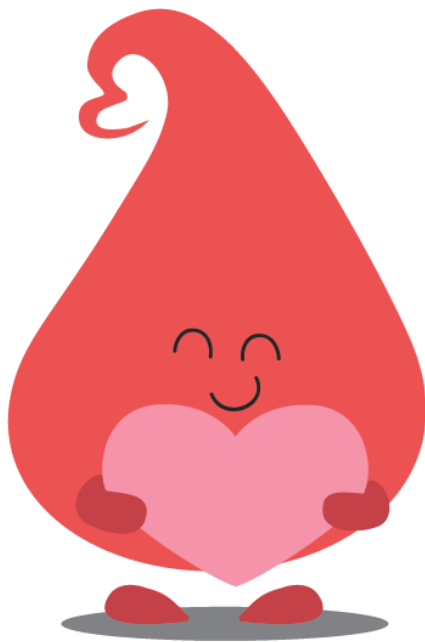
Home work (2):

Plot

$$1) x(t) = -4u(t + 4) - 3u(t + 1) + 5u(t - 1) + 2u(t - 3)$$

$$2) x(t) = 2r(t + 1) - 4r(t) + 4r(t - 1.5) - 2r(t - 2)$$

“Digital communication is the art of delivering meaning through noise and distance”



The End